

The Mechanisms of Muting: Deconstructing the Technology-Mediated Violence of Silence

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Abstract

This research addresses a critical gap in HCI: while the field engages with "harm," it inadequately conceptualizes "violence." One gap lies in how digital artifacts mediate structural violence through muting. Muting — the systemic silencing of marginalized groups — prevents vulnerable populations from accessing potentially life-saving resources and results in preventable morbidity and mortality. Drawing from Muted Group Theory, I demonstrate how technologies imbued with dominant values amplify muting in unprecedented ways through information suppression in suicide reporting, social-computing design that silences gender-based violence survivors, and epistemic inequity perpetuated by generative AI. My dissertation employs survivor-centered mixed methods — surveys, narrative interviews, and phenomenological analysis to understand how intimate partner violence survivors use digital artifacts in help-seeking. This work will produce the first empirical understanding of relationships between muting experiences and adverse outcomes, alongside design recommendations for remediating muting in help-seeking technologies. My goal is to establish cross-disciplinary approaches to violence prevention through ethical technology design.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

Keywords

Muting, Harm, Violence, Disclosure, Help-seeking, Prevention, Remediation, Social Computing

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1 Introduction & Research Questions

Harm remains a critical yet elusive concept—challenging to define, quantify, and address effectively. This ambiguity creates significant constraints within HCI and computing disciplines, potentially limiting (whether by design or inadvertently) our collective ability to fully confront the real-world consequences of technology design and implementation. There are frameworks that expand our understanding of harm [12], but existing work does not yet reveal how digital harms translate to public health outcomes. More importantly, at the time of this writing, no work in computing characterize this "harm" thoroughly as violence. The field's struggle with harm assessment impedes comprehensive, cross-disciplinary engagement with technology's adverse impacts on individuals and broader social systems.

Recognizing this, my work engages with muting, from Muted Group Theory [10], as a type of systematic silencing that constitutes structural violence. This theory posits that non-dominant groups are obstructed, excluded, or muted, by dominant groups from public discourse [10]. Public discourse, controlled by dominant groups, facilitates the social construction of identity. Consequently, non-dominant groups experience constraints in their self-advocacy and are forced to communicate within the norms of dominant groups [10] – muting. This ultimately limits help-seeking and prevention in potentially fatal ways – violence. I examine how digital artifacts influence this social phenomenon by mediating disclosure, self-understanding, and active help-seeking.

Though I have examined muting as violence in other contexts such as suicide [3, 5] and gender-based violence [4], my dissertation study centers survivors of intimate partner violence (IPV). While some work does examine survivor or supporter access to information and understanding of digital harm [6], it narrows on how digital artifacts are used against survivors to digitally mediate interpersonal harm [1, 9, 13] rather than systemic or structural harm. Thus, my thesis study intends survivor-centered research to quantify barriers to help-seeking and disclosure among IPV survivors toward understanding how said barriers relate to adverse outcomes. Additionally, I seek to characterize digital help-seeking behaviors to inform technology design. I intend to answer the following research questions:

- (1) How do survivors of IPV utilize digital technologies to divert experiences of muting?
- (2) Which experiences and perceptions of muting are predictors of adverse outcomes for survivors of IPV?
- (3) What can we learn about the implications of designing information-mediating technologies from survivor reflections on muting experiences and strategies for digital diversion?

2 The Mechanisms of Muting: Learnings from Prior Work

My completed work has indicated three mechanisms through which technologies mediate muting: removal of information, replication of social dynamics, and reconstruction of prejudice and partial knowledge.

2.1 Removal of Information

Digitally accessible venues of public discourse, such as news media, are avenues for at-risk populations to contextualize their experiences and hardships and inform behaviors such as help-seeking or isolation. In the case of suicide risk, the mechanism of removal of information obscures important sociodemographic details [5] and structural circumstances [3] in news reports of suicide. This ultimately results in vulnerable and marginalized groups not seeing themselves or their experiences represented in the collective experience and discussion of suicide. As a result, this can discourage help-seeking behavior due to increased stigma and isolation – limiting suicide prevention efforts.

2.2 Replication of Dynamics

In the context of gender-based violence, survivors face many challenges reporting their experiences to formal sources. Even if isolation and stigma are overcome, survivors face consequences of disclosure to professionals. As a result, they may turn to the affordances of pseudo-anonymity and online communities for support-seeking. The design of social computing platforms, however, may reinforce or replicate the social dynamics and power structures that enable muting offline [4].

2.3 Reconstruction of Prejudice and Partial Knowledge

Fundamentally, and more novel, survivors and at-risk populations may have a paucity of hermeneutic resources to even understand that they have experienced violence or abuse. These gaps in self-understanding challenge disclosure and help-seeking to formal sources likely due to gaps in available terminologies or concepts. Consequently, survivors may turn to tools such as generative AI in the form of large-language models to avoid muting offline and to understand their experiences. While these tools do afford access to relevant violence concepts, they also reconstruct offline epistemic inequities through disparate reading levels, masked by communication comprehension, and varying degrees of victim blaming and support (work in preparation).

Each of these mechanisms present with trade-offs, benefits and harms, to help-seeking populations. Despite these important findings, there still remains a key gap in understanding muting – remediation.

3 The Mechanism of Remediation: Digital Design to Mitigate Muting of Survivors of Intimate Partner Violence

Informed by the findings from prior work, my thesis work seeks to inform the practice of remediating mechanisms of digital muting in the context of the lived experience of violence. This study builds

on known barriers to disclosure and help-seeking experienced by survivors of intimate partner violence.

IPV is defined by the Centers for Disease Control and Prevention as “abuse or aggression that occurs in a romantic relationship” [2]. IPV impacts millions of people each year in the United States and often results in negative outcomes from chronic health conditions to death [7]. Survivors of IPV often face social and structural barriers when help-seeking, navigating critical resources such as health services and criminal justice intervention, and interacting with providers [11] –potentially increasing their risk of re-injury and even death as their needs go unaddressed. In the context of IPV, we can understand these barriers to disclosure as a secondary victimization of survivors manifesting as dismissal and indifference when talking about their experiences or being misdirected or inappropriately intervened upon when lacking the necessary hermeneutic resources to articulate experiences and seek support. It follows, then, that survivors may turn to technology to divert from these barriers.

3.1 Recruitment Strategy

This mixed-methods study employs purposive sampling to recruit survivors of IPV through two complementary pathways. First, I leverage established partnerships with service organizations in the metro-Atlanta area, primarily within the Grady Health System network, including the Injury Prevention and Research Center at Emory, the Grady Nia Project, the IVYY Project, and the Grady Trauma Recovery Center. These partnerships ensure participants have access to trauma-informed support throughout the research process. Second, I supplement clinical recruitment with outreach on social computing platforms where survivors self-identify, capturing experiences from those who may not have accessed formal support services.

3.2 Data Collection

I will collect data through two sequential methods. Initially, participants complete a comprehensive survey capturing demographic information, IPV experiences, muting incidents (barriers to help-seeking), adverse health and social outcomes, and digital diversion strategies. The survey incorporates validated measures of muting experiences adapted from systematic reviews of help-seeking barriers. Following survey completion, I conduct 15-25 semi-structured narrative interviews lasting up to 90 minutes with a subset of participants. These interviews employ a phenomenological approach, beginning with generative questions that encourage storytelling about navigating the information ecosystem during and after experiences of muting.

3.3 Analysis Plan

My analysis integrates quantitative and qualitative approaches. For survey data, I will employ descriptive statistics to characterize digital diversion practices and logistic regression models, as statistical power allows, to identify which muting experiences significantly predict adverse outcomes. Power analysis indicates a target sample of 300-500 participants is required to detect moderate effect sizes across multiple predictors. For interview data, I will use interpretive phenomenological analysis (IPA) [8] to examine how survivors

interpret their experiences and assign meaning to their navigation of information systems. This dual analytical approach allows us to quantify relationships between variables while preserving the richness of survivor narratives.

3.4 Progress

Though this work has not yet begun, the study is designed and partners are engaged. By the DC in October, I anticipate that survey data collection will be complete and narrative interviews with survey respondents will be meaningfully underway. This will be an optimal stage for valuable feedback as I plan my analysis to more clearly inform design recommendations and refine the tangible contributions of this work.

4 Expected Contributions

This research could fundamentally transform how we understand and support IPV survivors' information-seeking journeys. By uncovering the specific digital strategies survivors employ to overcome barriers to help, we can develop more responsive support systems tailored to their actual behaviors rather than ideal scenarios. Identifying which specific muting experiences most strongly predict negative outcomes enables practitioners across disciplines to prioritize interventions addressing the most harmful barriers. Perhaps most importantly, centering survivor voices in technology design could lead to a new generation of digital tools and platforms that actively counteract systemic silencing, creating pathways to support that work with—rather than against—survivors' natural coping mechanisms. These insights would bridge critical gaps between IPV support services, technology design, and public health approaches, potentially reducing help-seeking barriers that have remained stubbornly persistent despite decades of intervention efforts.

5 Benefits of the Doctoral Consortium

During my PhD working at the intersection of violence prevention and technology, I've found my research journey challenging but deeply meaningful. Coming from my background outside of computing, working over a decade in trauma-care, I faced insecurity about whether I belonged and if I had the right language to be understood. Establishing and sustaining a sense of community in computing is a top priority for me. First and foremost, participation in the DC extends that community immensely. I hope to develop career and life-long connections with brilliant people doing impactful work that continue to challenge and inspire me for years to come.

The DC comes at a critical juncture as I refine my approach to communicating my dissertation results. This timing is ideal for receiving diverse perspectives on my findings, improving how I articulate my arguments, and refining the conceptual framing of my contribution. I particularly seek guidance on identifying additional considerations for formulating design recommendations.

As I move towards the broader goal of shaping cross-disciplinary strategies for violence prevention, participation in the DC gives me the opportunity to put my work in front of a key audience that may utilize my findings in their own work. The collaborative environment would help me refine my ideas in ways that are most

useful for computing researchers. I'm excited to share my approach with the DC community and learn from others' experiences.

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